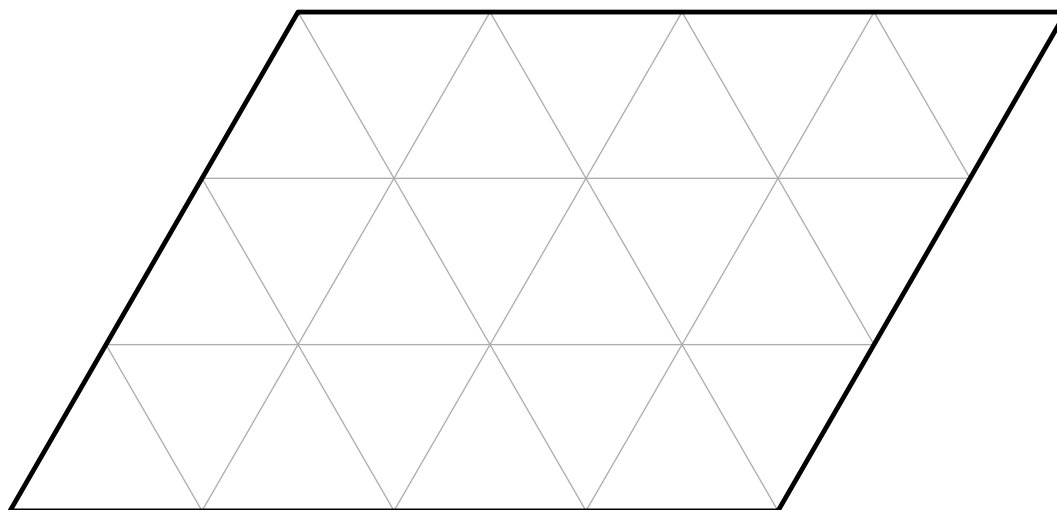


1 Ups and downs

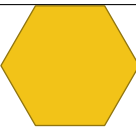
Name: _____

On our boards, every small triangle points **UP** ▲ or **DOWN** ▼.

1. Put a dot in every **UP** ▲ triangle on this board.



2. Put one block at a time on the board, lined up with the lines. How many ▲ and how many ▼ does it cover? Try different spots and turns!

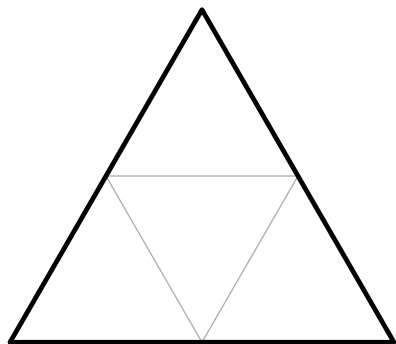
Block	covers ▲	covers ▼
		
		
		
		
		

Talk about it. Did any block give two different answers? Can you place a blue so that it covers two ▲ triangles?

2 Possible or impossible?

Cover a board with **blue rhombi only** : no gaps, no overlaps, nothing hanging over the edge. Circle **Possible** or **Impossible**. Then count the  and  triangles on the board.

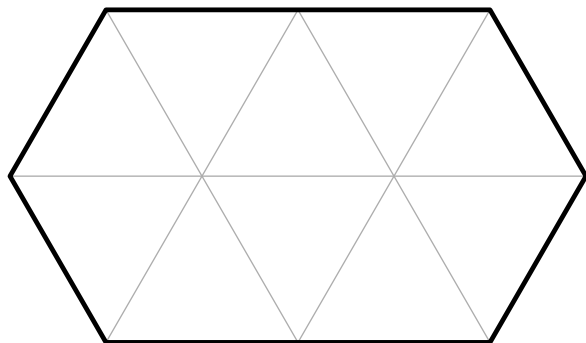
A



Possible Impossible

 _____  _____

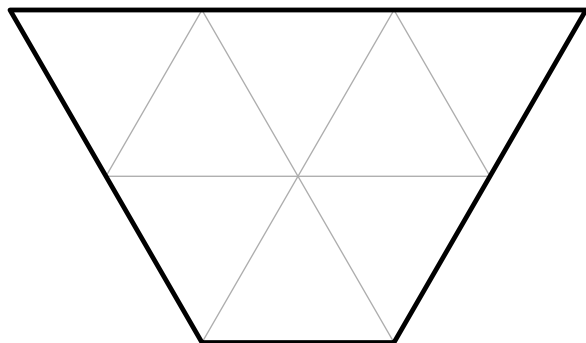
B



Possible Impossible

 _____  _____

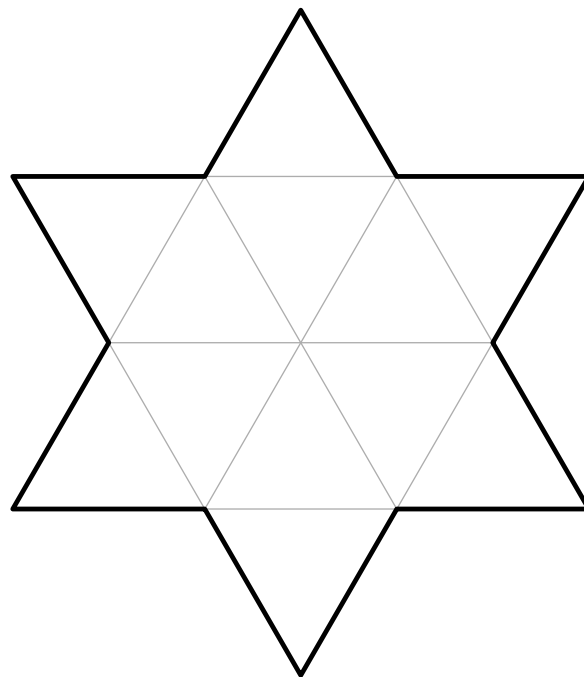
C



Possible Impossible

 _____  _____

D



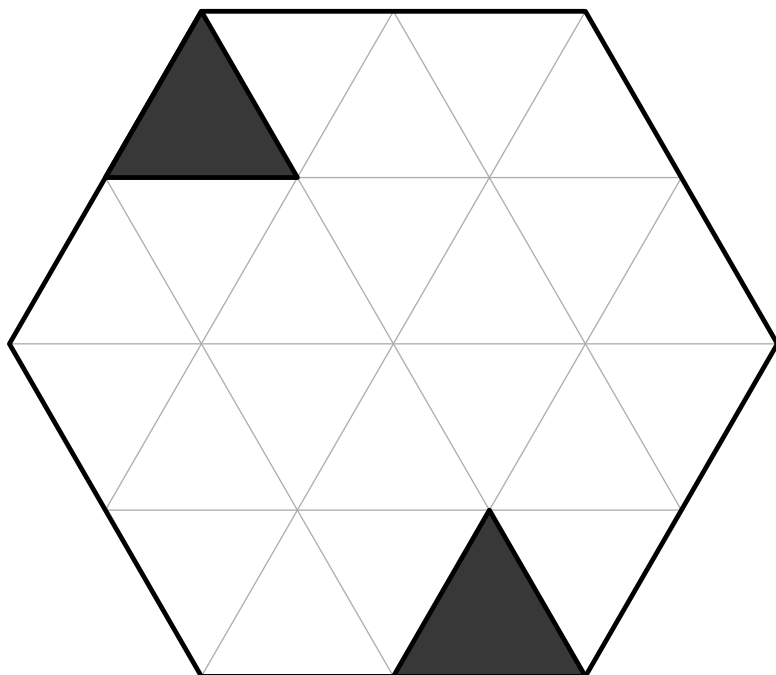
Possible Impossible

 _____  _____

Talk about it. Every board has an even number of triangles. What do the impossible boards have in common? Why does that make them impossible?

3 Broken hexagons

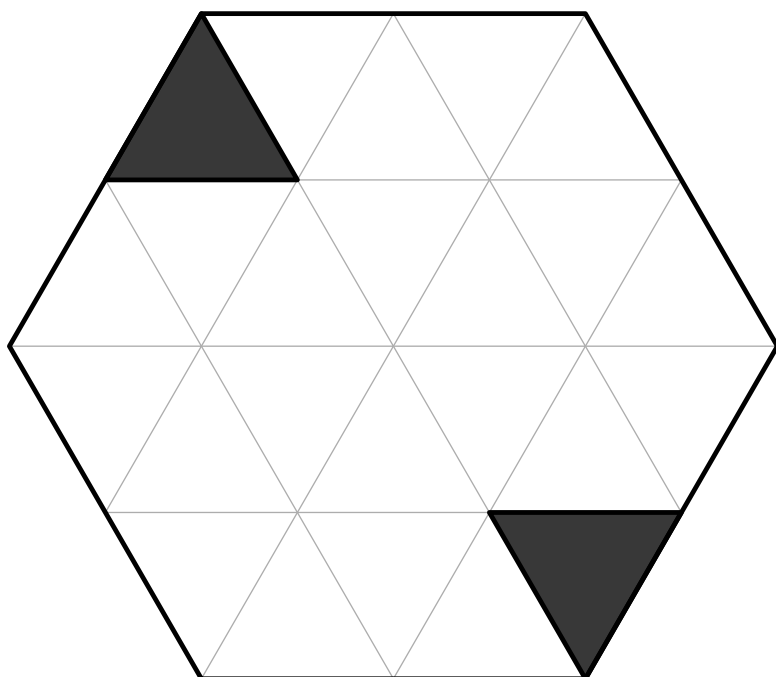
The black triangles are **holes**: no blocks go there. Try to cover all the white triangles with **blue rhombi only** .



X Hexagon X

Possible Impossible

 _____  _____



Y Hexagon Y

Possible Impossible

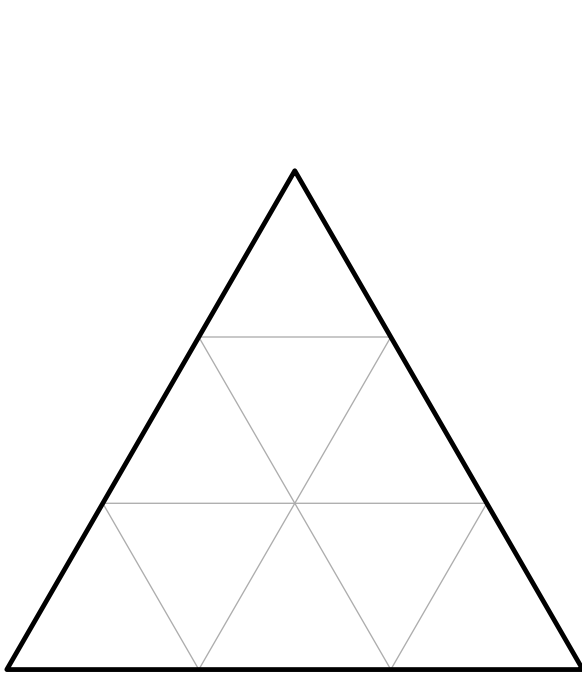
 _____  _____

Both hexagons have 22 white triangles. Why can one of them *never* be covered?

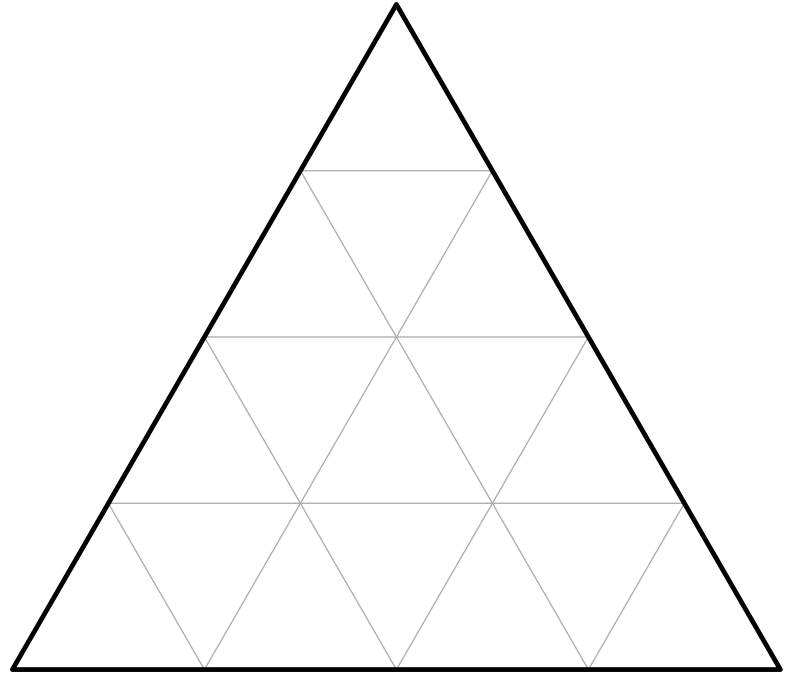
Make your own. Which two triangles could you take out of a whole hexagon so that it is *still* possible? So that it is impossible?

4 Fewest gaps

Can you cover these big triangles with blues alone? If not, put in as **many** blues  as you can, then fill the gaps with greens . Try to use as **few** greens as possible!



Side 3:  _____  _____



Side 4:  _____  _____

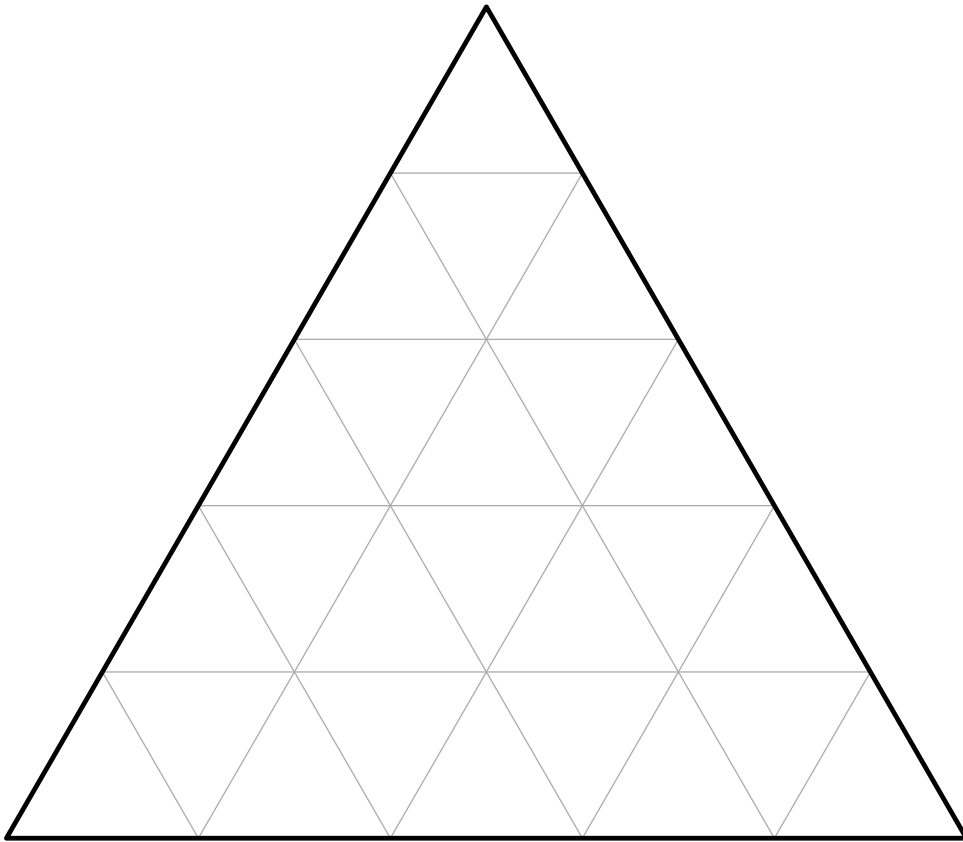
Your best so far. (Board A on page 2 is the side-2 triangle.)

Side	most blues 	fewest greens 
2		
3		
4		

Talk about it. Compare with your group. Can anyone beat your number of greens? Look at the triangles your greens are sitting on. Are they  or .

A plan that always works? Describe a way to place the blues that leaves only as many gaps as the side length. Would your plan work for *every* big triangle?

5 Bigger and bigger



Side 5. Use as many blues as you can. Fill the gaps with greens.



Purple puzzle. Would purple pieces help you leave fewer gaps? Try covering a purple with blues.

Fill in the table. Use your boards for sides 2–5. Can you *predict* sides 6 and 10?

Side	how many 	how many 	most blues 	fewest greens 
2				
3				
4				
5				
6				
10				

Why can nobody ever use fewer greens than this? (Hint: each blue covers one  and one .)
